Urban Restitution:

Sustainability has 3 metics Environment, Economy, and Equity.

Over the course of the mid to late 1900's urban planning in America has become increasingly focused on the individual rather than a sense of community as a whole. A result of this is a decreasingly friendly streetscape in many cities that aren't immediately dependent on tourism.

As such simply existing in a city has become increasingly about trying to get off the street and to your destination as soon as possible despite the fact that many great social experiences and safety systems stem from a lively streetscape.

Urban Restitution's goal is simply to act as a simple to use script that assists in discovering what interventions would make our streets better for people.

What it is

Urban Restitution is a Rhino Grasshopper script that is meant to input user provided environmental conditions regarding a site to provide a ranked list of intervention typologies that would best suit the site.

An additional secondary feature is a simple grid generator to further assist in the discovery process for where the typologies might be placed.

Urban Restitution is intended to be used by intro level designers or those who need an accelerated discovery process. It gives a direction to move in for further research.

What it does

Inputs:

- A pre-made .csv with environmental scores for each intervention.
- Simple to use sliders to adjust the environment being considered.
- Two simple geometries to act as the building and road interfaces.

Outputs:

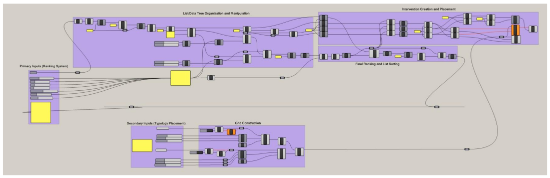
- A panel with each typology ranked alongside its total point score.
- A simple grid with simple geometries representing the highest ranked intervention placed.

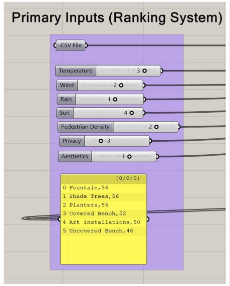
Methods

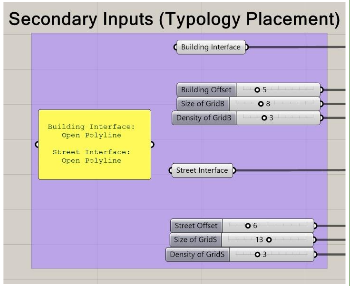
The bulk of the script is dedicated towards sorting the imported .csv file into a usable form and separating relevant info into certain areas.

With the data usable the scores for each intervention in the .csv are used alongside the multipliers from environmental condition sliders then added up to get an effective point score for each intervention and is then promptly sorted and returned to the user.

The input geometries are used to create offsets and from there a grid between the street and building, then data from the .csv is used to create a representative geometry of the highest ranked intervention on the grid.







XXXXXX	Covered Bench	Uncovered Bench	Shade Trees	Fountain	Planters	Art installations
Round	0	0	1	1	0	1
W	3	3	10	4	5	4
H	1	1	10	4	2	4
Temp+	4	3	5	3	3	3
Temp-	2	4	1	3	2	3
Wind+	2	1	3	3	2	2
Wind-	4	5	2	3	3	4
Rain+	2	1	4	2	2	2
Rain-	4	5	3	4	3	4
Sun+	5	3	5	4	5	3
Sun-	2	3	1	3	2	3
Pedestrian_Density+	3	4	2	3	3	3
Pedestrian_Density-	4	3	4	3	3	4
Privacy+	4	2	4	2	4	2
Privacy-	2	4	1	4	3	4
Aesthetics+	2	2	4	5	5	5
Aesthetics-	3	3	2	2	2	1

Results

When it comes to the functionality of Urban Restitution there is an overall success.

It is capable of providing the ranked list as well as placing representations of the interventions on a simple site.

Its greatest strength happens to also be a potential weakness. Being incredibly simple means it is very user friendly with minimal inputs and few possible unforeseen circumstances. It is also very easy to expand in terms of the database of interventions and environmental conditions.

However it is going to be stuck with a very limited use case without the addition of other features further down the line.

Challenges

A potential issue with this program is that due to its simplicity there is always going to be an issue of nice/rare use cases which would not fall close enough to any specified typing to give accurate results. The results themselves may also begin to fall flat when considered under higher layers of scrutiny.

A major difficulty Urban Restitution is likely to encounter is building codes. Due to high levels of specificity and variability between cities' codes having a single database and giving appropriate recommendations could prove to be ineffective unless done at larger than expected scales.